

Degrader-Antibody Conjugates Targeting c-Rel in Pathogenic T Cells for Treatment of Autoimmune and Inflammatory Diseases

Executive Summary

We propose a novel Degrader-Antibody Conjugate (DAC) therapeutic platform designed to selectively degrade c-Rel, a critical intracellular transcription factor driving pathogenic T cell function in autoimmune diseases. The DAC utilizes CD25 (IL-2 receptor α subunit) targeting for selective internalization into activated T cells, followed by intracellular release of a PROTAC-based degrader payload. This approach combines the precision of antibody-mediated targeting with the catalytic efficiency of targeted protein degradation to achieve high specificity, sustained therapeutic effect, and minimal off-target toxicity.

Primary Indication: Systemic Lupus Erythematosus (SLE)

Secondary Indications: Psoriasis, Crohn's Disease, Ulcerative Colitis

1. Scientific Rationale

1.1 Target Selection: c-Rel

c-Rel is a member of the NF- κ B/Rel family of transcription factors and serves as a master regulator of pathogenic T cell differentiation and function. Key characteristics supporting its therapeutic targeting include:

- **Disease-specific role:** c-Rel is selectively required for differentiation, proliferation, and cytokine production of effector Th1, Th17, and Tfh cells implicated in lupus, psoriasis, and inflammatory bowel disease
- **Non-redundant function:** Unlike NF- κ B p65, c-Rel has distinct, non-overlapping roles in autoimmune pathogenesis, reducing concerns about compensatory mechanisms
- **Preclinical validation:** c-Rel knockout mice demonstrate resistance to experimental autoimmune encephalomyelitis (EAE), lupus, colitis, and psoriasis models
- **Safety profile:** c-Rel is non-essential in most healthy tissues, offering a favorable therapeutic window

1.2 Delivery Vehicle: CD25-Targeted Internalization

CD25 (IL-2 receptor α chain) serves as the extracellular targeting antigen with the following advantages:

- **Selective expression:** Upregulated on activated T cells and regulatory T cells in autoimmune tissue environments
- **Internalization competency:** Well-established receptor-mediated endocytosis upon IL-2 engagement

- **Clinical precedent:** CD25-targeting antibodies (basiliximab, daclizumab) have demonstrated clinical safety in transplant medicine
- **Efficient payload delivery:** CD25-targeting antibody-drug conjugates have shown reproducible internalization and payload release

2. DAC Design and Components

2.1 Molecular Architecture

The DAC consists of four primary components:

1. **Antibody:** Humanized anti-CD25 monoclonal antibody (Fc-engineered IgG1)
2. **Linker:** Cleavable PEGylated linker (cathepsin B- or acid-sensitive)
3. **Degrader payload:** c-Rel ligand conjugated to E3 ligase recruiter (PROTAC-style)
4. **Endosome escape motif:** Fusogenic sequence or pH-triggered release mechanism

2.2 Mechanism of Action

The DAC operates through the following sequential steps:

1. **Binding:** Anti-CD25 antibody binds to IL-2R α on activated pathogenic T cells
2. **Internalization:** DAC undergoes receptor-mediated endocytosis via IL-2R pathway
3. **Payload release:** Acid-sensitive linker cleaves in endosomal compartment
4. **Cytoplasmic delivery:** Endosome escape motif facilitates degrader release into cytoplasm
5. **Target engagement:** PROTAC warhead simultaneously binds c-Rel and recruits E3 ligase (VHL or CRBN)
6. **Protein degradation:** c-Rel undergoes ubiquitination and proteasomal degradation
7. **Therapeutic effect:** Loss of c-Rel function leads to suppression of pathogenic T cell responses

3. Therapeutic Advantages

3.1 Comparison to Conventional Therapies

Therapy Class	Limitations	DAC Solution
Monoclonal antibodies	Extracellular targets only	Intracellular transcription factor targeting
Oral immunosuppressants	Systemic toxicity, broad effects	T cell-specific delivery
JAK/STAT inhibitors	Broad inhibition, long-term risks	Modular, lower-dose mechanism
Corticosteroids	Non-selective, significant side effects	Precision degradation

3.2 Key Advantages

- **Catalytic activity:** Substoichiometric degrader activity enables lower dosing requirements
- **Sustained effect:** Protein degradation provides longer-lasting therapeutic benefit compared to inhibition
- **Minimal systemic exposure:** Targeted delivery reduces off-target effects
- **Modular design:** Platform amenable to targeting other intracellular drivers
- **Disease specificity:** Selective targeting of pathogenic T cells while preserving normal immune function

4. Development Strategy

4.1 Phase 1: Discovery and Optimization

Antibody Selection:

- Engineer humanized anti-CD25 antibody from clinically validated scaffolds
- Optimize binding affinity and internalization kinetics
- Fc engineering for enhanced stability and reduced immunogenicity

Degrader Design:

- Synthesize c-Rel-targeting PROTAC with validated ligands
- Screen E3 ligase recruiters (VHL, CRBN, MDM2) for optimal degradation efficiency
- Optimize linker length and composition for maximal activity

Linker Development:

- Design pH-cleavable PEG linkers for controlled endosomal release
- Incorporate endosome escape motifs for efficient cytoplasmic delivery
- Evaluate stability in plasma and target tissues

4.2 Phase 2: In Vitro Validation

Target Engagement Studies:

- Confirm CD25 expression and internalization kinetics on primary human T cells
- Validate c-Rel degradation via Western blot and mass spectrometry
- Assess dose-response relationships and kinetics of degradation

Functional Assays:

- Measure suppression of pathogenic cytokines (IL-17, IFN- γ , GM-CSF)
- Evaluate effects on T cell proliferation and apoptosis
- Assess selectivity for activated vs. resting T cells

4.3 Phase 3: Preclinical Efficacy Models

Disease Models:

- MRL/lpr lupus mice for SLE indication
- DSS-induced colitis model for IBD
- Imiquimod-induced psoriasis model

Efficacy Endpoints:

- Disease activity scores (SLEDAI for lupus)
- Histological assessment of target organ damage
- Cytokine profiling and immune cell phenotyping
- Pharmacokinetic and biodistribution studies

4.4 Phase 4: IND-Enabling Studies

Safety Assessment:

- Comprehensive toxicology studies in non-human primates
- Immunogenicity evaluation
- Reproductive and developmental toxicity studies

Manufacturing Development:

- Scalable conjugation process development
- Analytical method development and validation
- Formulation optimization for clinical use

5. Biomarkers and Monitoring

5.1 Target Engagement Biomarkers

- **c-Rel protein levels:** Western blot and proteomics analysis
- **c-Rel mRNA expression:** qRT-PCR analysis
- **CD25 receptor occupancy:** Flow cytometry assessment

5.2 Pharmacodynamic Biomarkers

- **Cytokine suppression:** IL-17, IFN- γ , GM-CSF levels
- **T cell activation markers:** CD69, CD25, HLA-DR expression
- **Transcriptional signatures:** RNA-seq analysis of T cell subsets

5.3 Safety Biomarkers

- **Liver function:** ALT, AST, bilirubin
- **Renal function:** Creatinine, BUN
- **Hematological parameters:** Complete blood count with differential
- **Cytokine storm markers:** IL-6, TNF- α , IL-1 β

6. Platform Expandability

6.1 Additional Targets

The DAC platform can be extended to target other intracellular drivers of autoimmune disease:

- **STAT3:** Th17 differentiation and function
- **IRF4:** Plasma cell differentiation and antibody production
- **ROR γ t:** Th17 lineage specification
- **T-bet:** Th1 differentiation and IFN- γ production

6.2 Alternative Cell Types

- **B cells:** Anti-CD22 or anti-CD19 targeting for autoantibody-producing cells
- **Macrophages:** Anti-CSF1R targeting for inflammatory macrophages
- **Dendritic cells:** Anti-DEC205 targeting for antigen-presenting cells

6.3 Oncology Applications

- **Regulatory T cells:** Degradation of FoxP3 or STAT5 in tumor microenvironment
- **Tumor-associated macrophages:** Targeting of pro-tumorigenic transcription factors

7. Regulatory and Commercial Considerations

7.1 Regulatory Pathway

- **FDA designation:** Potential for Fast Track or Breakthrough Therapy designation
- **Orphan drug status:** Possible for rare autoimmune indications

- **Biomarker qualification:** Collaboration with FDA on companion diagnostics

7.2 Intellectual Property

- **Platform IP:** Core DAC technology and delivery mechanisms
- **Target-specific IP:** c-Rel degraders and CD25-targeting antibodies
- **Method of use:** Therapeutic applications in autoimmune diseases

7.3 Market Opportunity

- **Lupus market:** \$2.3 billion globally with significant unmet need
- **Psoriasis market:** \$12.4 billion with growing biologics segment
- **IBD market:** \$6.8 billion with demand for precision therapies

8. Risk Assessment and Mitigation

8.1 Technical Risks

- **Degrader efficacy:** Mitigation through systematic PROTAC optimization
- **Internalization efficiency:** Alternative targeting antigens identified
- **Payload stability:** Comprehensive linker chemistry evaluation

8.2 Regulatory Risks

- **Novel modality:** Close collaboration with regulatory agencies
- **Safety profile:** Comprehensive preclinical safety package
- **Manufacturing complexity:** Early CMC development engagement

8.3 Commercial Risks

- **Competition:** Differentiated mechanism of action and superior efficacy
- **Market access:** Health economic studies and real-world evidence generation
- **Manufacturing cost:** Process optimization and scale-up strategies

9. Timeline and Milestones

Year 1-2: Discovery and Optimization

- Lead DAC selection and characterization
- In vitro proof-of-concept studies
- IND-enabling study initiation

Year 3-4: Preclinical Development

- Efficacy studies in disease models
- Safety assessment in non-human primates
- Manufacturing process development

Year 5-6: Clinical Development

- Phase 1 first-in-human studies
- Dose escalation and preliminary efficacy
- Biomarker validation

Year 7-8: Proof-of-Concept

- Phase 2a studies in lupus patients
- Expansion to additional indications
- Platform extension to new targets

10. Conclusion

The proposed DAC therapeutic platform represents a paradigm shift in autoimmune disease treatment by enabling selective intracellular targeting of disease-driving transcription factors. By combining the precision of antibody-mediated targeting with the catalytic efficiency of protein degradation, this approach offers the potential for superior efficacy with improved safety compared to conventional immunosuppressive therapies.

The c-Rel-targeting DAC for lupus and other autoimmune diseases addresses significant unmet medical needs while establishing a modular platform for future therapeutic development. With strong scientific rationale, clear development pathway, and substantial market opportunity, this program represents an attractive investment in next-generation precision medicine for autoimmune diseases.

This document represents a comprehensive therapeutic development strategy for Degradable-Antibody Conjugates targeting c-Rel in autoimmune diseases. The proposed approach leverages cutting-edge protein degradation technology with established antibody delivery mechanisms to create a novel therapeutic modality with significant clinical and commercial potential.